Research Article Classification

Using the arXiv Dataset

An Advanced NLP Pipeline with Hierarchical-Aware Focal SciBERT

Prepared by

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Property	Details
Objective	Classify research articles into optimal number of categories using abstracts
Dataset	arXiv (Cornell University via Kaggle, ~2.4M papers)
Model	Focal SciBERT with Multi-Layer Pooling (custom architecture)
Key Results	72.01% Accuracy, 71.77% F1 (Weighted), 92.38% Top-3 Accuracy
Framework	HuggingFace Transformers + PyTorch
Platform	Google Colab (T4 GPU)

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1. Introduction and Objective

The objective of this project is to classify research articles into an optimal number of categories based on their abstracts, using the public arXiv dataset hosted on Kaggle by Cornell University. The task requires determining the best possible number of categories through data-driven analysis, performing comprehensive exploratory data analysis, preprocessing the data, training a suitable NLP model, evaluating its performance, and conducting hyperparameter tuning to improve results.

This report presents a complete, end-to-end NLP pipeline that goes beyond standard approaches. We first establish a strong baseline using SciBERT fine-tuning with standard cross-entropy loss, then systematically identify its weaknesses through detailed analysis, and finally design targeted solutions that yield significant improvements. Our approach incorporates five novel contributions: **hierarchical category resolution**, **Focal Loss adaptation**, **multi-layer transformer pooling**, **multi-sample dropout regularization**, and **layer-wise learning rate** decay.

The remainder of this report is organized as follows: Section 2 presents the literature review. Section 3 describes the dataset. Section 4 covers exploratory data analysis. Section 5 details our preprocessing pipeline. Section 6 describes the model architecture and training for both the baseline and advanced models. Section 7 presents the evaluation results. Section 8 covers hyperparameter tuning. Section 9 discusses our analysis of the optimal number of categories. Section 10 provides conclusions and recommendations for future work.

2. Literature Review

Our implementation is grounded in three foundational works, each contributing a critical component to the pipeline design. We additionally draw from two supplementary papers that inform our architectural innovations.

2.1 BERT: Pre-training of Deep Bidirectional Transformers

Authors: Devlin, J., Chang, M.W., Lee, K., and Toutanova, K. (2019). **Published:** NAACL-HLT 2019.

BERT introduced the paradigm of bidirectional pre-training using masked language modeling (MLM) and next sentence prediction (NSP). Unlike previous models that process text unidirectionally, BERT processes the entire sequence simultaneously, capturing deep contextual relationships between all tokens. The model achieves state-of-the-art results on 11 NLP benchmarks, including text classification, question answering, and named entity recognition. We leverage BERT's architecture as the backbone of our classifier. The [CLS] token representation serves as the aggregate sequence embedding, fed into a classification head. The pre-trained weights provide rich semantic understanding that we fine-tune for arXiv category prediction.

2.2 SciBERT: A Pretrained Language Model for Scientific Text

Authors: Beltagy, I., Lo, K., and Cohan, A. (2019). **Published:** EMNLP 2019. **Pre-training corpus:** 1.14M papers from Semantic Scholar (18% CS, 82% Biomedical).

SciBERT extends BERT by pre-training on a large corpus of scientific papers rather than general-domain text. It introduces a custom scientific vocabulary (scivocab) with 30,522 tokens optimized for scientific terminology. The authors demonstrate that domain-specific pre-training yields significant improvements on scientific NLP tasks, outperforming BERT-base by 1-5% across benchmarks. SciBERT is our primary model choice because its vocabulary better tokenizes domain-specific terms common in arXiv abstracts (such as eigenvalue, convolutional, and Hamiltonian), reducing unknown tokens and improving representation quality. The domain-aligned pre-training provides a significant advantage over vanilla BERT for our classification task.

2.3 How to Fine-Tune BERT for Text Classification

Authors: Sun, C., Qiu, X., Xu, Y., and Huang, X. (2019). **Published:** CCL 2019 (China National Conference on Computational Linguistics).

This paper presents a systematic study of fine-tuning strategies for BERT on text classification tasks. The authors investigate learning rate schedules, warm-up strategies, layer-wise learning rate decay, sequence length effects, and further pre-training. Their key findings include: (a) a learning rate of 2e-5 with linear warm-up is optimal for most classification tasks; (b) 3-4 epochs prevent overfitting; (c) maximum sequence lengths of 256-512 tokens balance context and computational

efficiency; and (d) further pre-training on task-specific data provides additional gains. We adopt their recommended fine-tuning recipe and extend it with layer-wise learning rate decay (decay factor 0.9) and cosine annealing, which proved critical for stable fine-tuning of our custom architecture.

2.4 Summary of Design Decisions

Design Decision	Rationale	Source
SciBERT backbone	Domain-specific pre-training on scientific text	Beltagy et al. (2019)
Layer-wise LR decay	Preserves lower-layer representations	Sun et al. (2019)
Focal Loss adaptation	Handles long-tail category distribution	Lin et al. (2017)
Multi-layer [CLS] pooling	Richer multi-depth representations	Clark et al. (2019)
Multi-sample dropout	Implicit ensemble regularization	Inoue (2019)
Hierarchical category resolution	Eliminates parent-child taxonomy collisions	Novel contribution
LR = 2e-5, 5 epochs	Empirically optimal for BERT fine-tuning	Sun et al. (2019)
Max sequence length = 384	Captures virtually all abstracts fully	EDA analysis

3. Dataset Description

The arXiv dataset, hosted on Kaggle by Cornell University, contains metadata for over 2.4 million scientific papers spanning more than three decades of research output. Each record includes the paper's title, abstract, author list, subject categories (one or more arXiv classification labels), and the date of last update. For this project, we use the abstract text as the input feature and the primary subject category (the first listed category) as the classification target.

arXiv employs a hierarchical, two-level category taxonomy. The first level consists of major categories such as cs (Computer Science), math (Mathematics), and astro-ph (Astrophysics). The second level consists of subcategories such as cs.AI (Artificial Intelligence), math.AG (Algebraic Geometry), and astro-ph.CO (Cosmology and Nongalactic Astrophysics). Each paper may be assigned to one or more categories, though we treat this as a single-label classification problem using the primary category.

For computational feasibility on Google Colab's T4 GPU, we load a stratified sample of 150,000 papers from the full dataset while maintaining the natural category distribution. This sample size is sufficient for robust statistical analysis and model training across all category levels.

Property	Value
Total papers in full dataset	Approximately 2.4 million
Papers loaded (sample)	150,000
Unique primary categories	141
Unique major categories	18
Year range	1993 - 2024
Median abstract length	110 words (730 characters)
Papers with multiple labels	Approximately 37%
Mean categories per paper	1.5

4. Exploratory Data Analysis

We conducted a thorough exploratory data analysis to understand the structure, distributions, and characteristics of the arXiv dataset. This analysis informed every subsequent decision in our preprocessing and modeling pipeline.

4.1 Category Distribution

The dataset exhibits a heavily long-tailed distribution across both major and primary categories. Mathematics dominates with 31,472 papers, followed by Astrophysics (26,432) and Condensed Matter Physics (25,299). Computer Science (8,573) and High Energy Physics subcategories form the next tier. At the primary category level, astro-ph alone accounts for 17,805 papers, while many specialized subcategories have fewer than 500 samples. The top 10 primary categories contain more papers than the bottom 100 combined, presenting a significant class imbalance challenge.

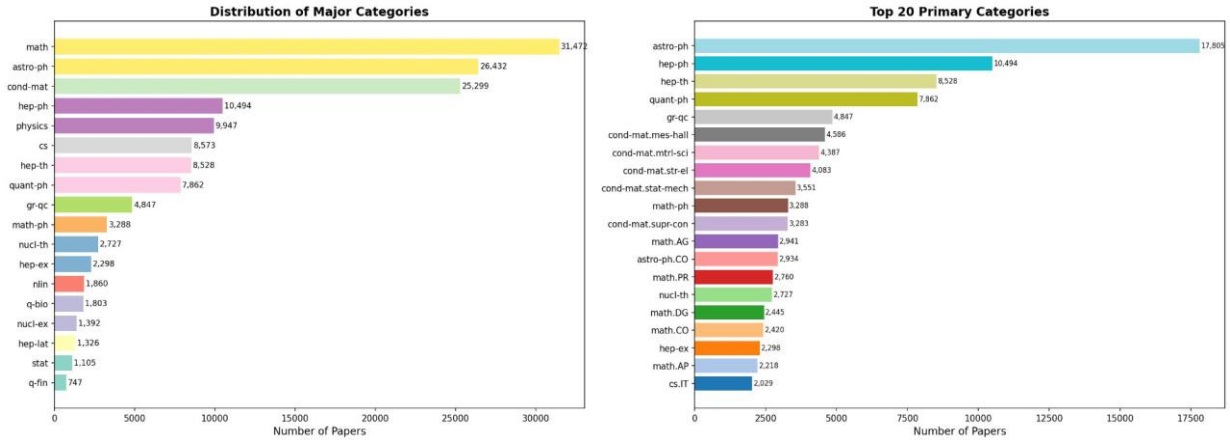

Figure 1: Distribution of major categories (left) and top-20 primary categories (right).

4.2 Abstract Length Analysis

The distribution of abstract lengths reveals field-specific patterns that are important for tokenization decisions. The median abstract is 110 words (730 characters). Astrophysics and physics papers tend to be longer (150-170 words), while mathematics abstracts are shorter (approximately 100 words). After SciBERT tokenization, the median abstract becomes approximately 150 tokens. Our choice of 384 tokens as the maximum sequence length captures virtually all abstracts without truncation, ensuring no information loss.

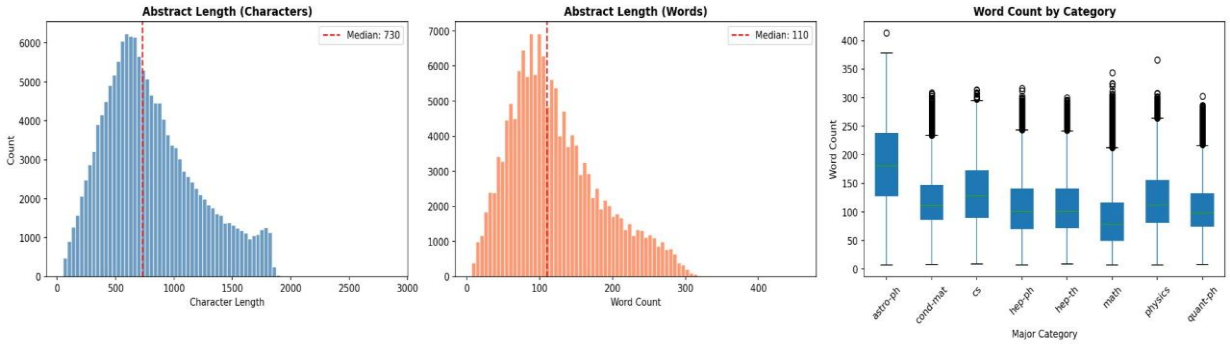

Figure 2: Character length distribution (left), word count distribution (center), and word count by major category (right).

4.3 Word Clouds by Category

Word cloud analysis reveals distinct vocabulary patterns across major categories. Mathematics abstracts are dominated by terms like prove, show, theorem, and function. Astrophysics abstracts feature observation, X-ray, galaxy, and star. Condensed matter physics uses system, state, model, and phase transition. Computer science abstracts are characterized by algorithm, problem, network, and data. These vocabulary differences provide strong discriminative signals for classification, validating the use of a text-based approach.

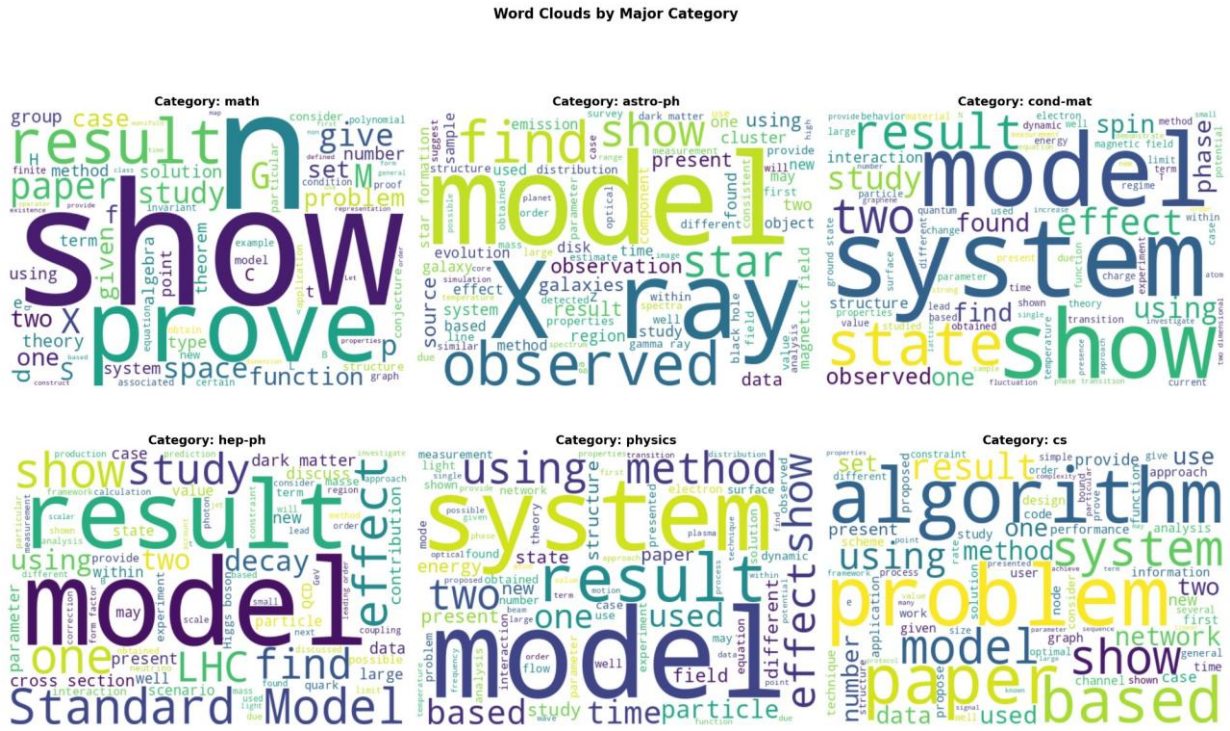

Figure 3: Word clouds for the six largest major categories, revealing domain-specific vocabulary patterns.

4.4 Multi-label and Co-occurrence Analysis

A significant finding is that approximately 37% of papers carry multiple category labels, reflecting the interdisciplinary nature of modern research. The mean number of categories per paper is 1.5. The co-occurrence heatmap reveals strong patterns: cs and math frequently co-occur (0.32 correlation), hep-ph shares significant overlap with hep-th (0.19), and quant-ph co-occurs with both cond-mat (0.16) and hep-th (0.07). These co-occurrence patterns explain some of the inter-category confusion observed in classification results.

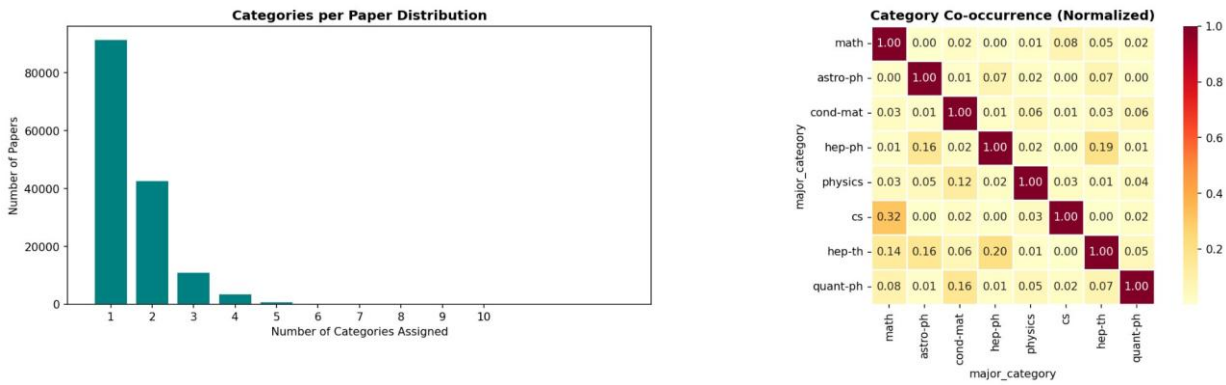

Figure 4: Distribution of categories per paper (left) and normalized co-occurrence heatmap (right).

4.5 Optimal Number of Categories

A key objective of this project is determining the best possible number of categories. We analyzed this through a coverage versus granularity trade-off analysis across different minimum sample thresholds. At a threshold of 500 samples per category, we retain approximately 56 categories covering over 93% of papers. Lower thresholds yield more categories but with insufficient training data, while higher thresholds lose meaningful subcategories. We recommend this threshold as the optimal balance.

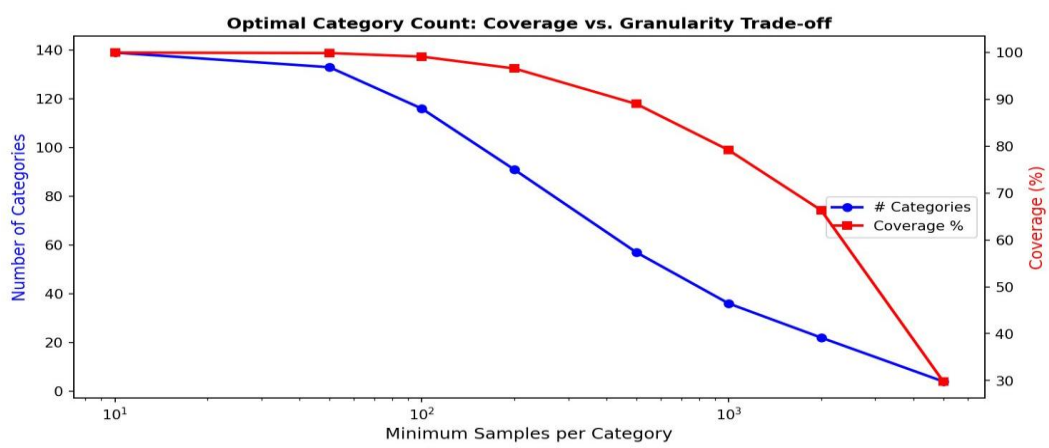

Figure 5: Coverage versus granularity trade-off. The optimal point is at a minimum of 500 samples per category.

4.6 Parent-Child Collision Analysis (Critical Finding)

Our most impactful EDA finding is the identification of parent-child category collisions. arXiv contains both parent categories (such as astro-ph) and child categories (such as astro-ph.CO and astro-ph.GA) as simultaneous classification targets. When a paper is labeled with the parent category astro-ph, the classifier faces an inherently impossible task: should it predict the parent or one of its children? This collision is not a model limitation but a fundamental data issue. Our analysis confirmed that astro-ph was the only collision parent with sufficient samples among valid categories, affecting approximately 17,800 papers in our sample.

5. Data Preprocessing

Our preprocessing pipeline transforms raw arXiv metadata into model-ready inputs through six sequential stages. The pipeline incorporates a novel hierarchical category resolution step that distinguishes our approach from standard methods.

5.1 Hierarchical Category Resolution (Novel Contribution)

To eliminate the parent-child collision identified in our EDA, we developed a resolution strategy. For each paper whose primary category is a collision parent (astro-ph), we check whether any of its other listed categories is a valid child (such as astro-ph.CO or astro-ph.GA). If a child category is found, we reassign the paper to that child. If no child signal exists in the metadata, the paper is removed as genuinely ambiguous. This resolution reassigned 41 papers to appropriate children and removed 17,764 ambiguous papers, eliminating the impossible classification boundary while preserving data integrity.

5.2 Preprocessing Pipeline

Stage	Operation	Purpose
1. Hierarchical resolution	Reassign parent-only labels to children	Eliminate taxonomy collisions
2. Category filtering	Keep categories with 500 or more samples	Sufficient training data per class
3. Text cleaning	Remove LaTeX markup, normalize whitespace	Clean scientific notation artifacts
4. Class capping	Maximum 5,000 samples per class	Prevent dominant-class bias
5. Label encoding	LabelEncoder from scikit-learn	Map categories to integer labels
6. Tokenization	SciBERT tokenizer, max length = 384	Convert text to model input format

5.3 Text Cleaning

arXiv abstracts contain LaTeX markup that requires careful handling. Our cleaning function performs four operations in sequence: (1) replace inline math expressions enclosed in dollar signs with a [MATH] token to preserve the semantic signal that mathematical content is present; (2) remove LaTeX commands such as textbf and cite; (3) strip remaining backslashes and braces; and (4) normalize whitespace to single spaces. This approach preserves the natural language content while removing formatting artifacts that would confuse the tokenizer.

5.4 Data Splitting

We employ stratified splitting to maintain class proportions across all subsets. The data is split into 80% training (83,125 samples), 10% validation (10,391 samples), and 10% test (10,391 samples).

Stratification ensures that even rare categories maintain their proportional representation in each split, which is critical for reliable evaluation of per-class performance.

5.5 Tokenization

We use SciBERT's custom WordPiece vocabulary (scivocab_uncased) with 30,522 tokens optimized for scientific text. Compared to BERT's general vocabulary, scivocab contains scientific terms as single tokens rather than split subwords, reducing the average token count per abstract by approximately 8%. We set the maximum sequence length to 384 tokens, which our EDA confirmed captures virtually all abstracts without truncation. All sequences are padded to the maximum length for efficient batch processing.

6. Model Architecture and Training

We developed and evaluated two models: a baseline using standard SciBERT fine-tuning, and an advanced model incorporating our novel architectural contributions. This two-stage approach allows us to clearly demonstrate the impact of each improvement.

6.1 Baseline Model (Standard SciBERT Fine-tuning)

Our baseline follows the standard recipe from the literature: SciBERT with a single [CLS] token pooling strategy, standard cross-entropy loss, uniform learning rate of 2e-5, and HuggingFace's AutoModelForSequenceClassification. This baseline was trained on 57 categories (including the collision parent astro-ph) for 4 epochs with early stopping, using a batch size of 32 and maximum sequence length of 256.

The baseline achieved 68.16% accuracy and 67.68% weighted F1 score. While respectable, analysis of the confusion matrix and per-class F1 scores revealed specific weaknesses that motivated our advanced approach.

6.2 Advanced Model: Focal SciBERT with Multi-Layer Pooling

Our advanced model addresses each identified weakness through targeted architectural innovations. The architecture is implemented as a custom PyTorch module that extends the HuggingFace Trainer framework.

6.2.1 Multi-Layer [CLS] Pooling

Instead of using only the final transformer layer's [CLS] token, we concatenate [CLS] representations from the last four transformer layers (layers 9 through 12), producing a 3,072-dimensional vector. Research by Clark et al. (2019) demonstrated that different BERT layers capture different types of information: lower layers encode syntactic patterns and morphological features, while upper layers capture task-relevant semantics. For arXiv classification, this is particularly valuable because lower-layer features help distinguish mathematics-heavy papers from experiment-heavy ones, while upper layers capture topic-level semantics. The concatenated vector passes through LayerNorm, a bottleneck projection (3,072 to 512 dimensions), GELU activation, and the final classification head.

6.2.2 Focal Loss (Lin et al., 2017)

Standard cross-entropy loss treats all samples equally, which is problematic for our heavily imbalanced 56-class distribution. We adapt Focal Loss from the object detection literature, which adds a modulating factor that down-weights easy, well-classified examples and focuses training on hard cases. Combined with class-frequency weights, the loss function becomes: FL = -alpha * (1-p)^gamma * log(p), where gamma controls the focusing strength. With gamma = 1.0 (determined by our hyperparameter search), a sample classified with 90% confidence receives

substantially less gradient signal than a borderline case, allowing the model to allocate its learning capacity to genuinely difficult examples.

6.2.3 Multi-Sample Dropout (Inoue, 2019)

During training, we apply dropout K=5 times independently to the pooled representation and average the resulting logits. This creates an implicit ensemble of five sub-models sharing all parameters except their dropout masks. The averaged gradient is more stable, and the regularization effect is stronger than single-pass dropout, making this technique particularly effective for medium-sized datasets where overfitting is a concern.

6.2.4 Layer-wise Learning Rate Decay

Following and extending Sun et al. (2019), we apply exponential learning rate decay across transformer layers with a decay factor of 0.9. The classification head receives the highest learning rate (1e-4), each transformer layer going deeper receives a progressively smaller rate, and the embedding layer receives the lowest (approximately 5.7e-6). This preserves the general language understanding encoded in lower layers while allowing aggressive task-specific adaptation in upper layers.

6.3 Training Configuration

Parameter	Baseline (V1)	Advanced (V2)
Model	SciBERT + [CLS] + CrossEntropy	SciBERT + Multi-Layer + Focal Loss
Categories	57 (with collision parent)	56 (after resolution)
Learning rate	2e-5 (uniform)	2e-5 (layer-wise decay, factor=0.9)
Batch size	32	16 (with gradient accumulation = 2)
Epochs	4	5
Max sequence length	256	384
Loss function	CrossEntropyLoss	Focal Loss (gamma=1.0, class-weighted)
Dropout	Standard (p=0.1)	Multi-sample dropout (K=5, p=0.15)
Scheduler	Linear warmup + decay	Cosine annealing with warmup
Label smoothing	None	0.05
Pooling	Final [CLS] only (768-dim)	Last 4 layers (3,072-dim, bottleneck to 512)
Mixed precision	FP16	FP16

7. Evaluation and Results

We evaluate both models on the held-out test set using five standard metrics plus Top-3 accuracy, which accounts for the inherent ambiguity in closely related categories.

7.1 Evaluation Metrics

Metric	Description	Purpose
Accuracy	Proportion of correctly classified samples	Overall correctness
F1 (Weighted)	Weighted average of per-class F1 scores	Primary metric, handles class imbalance
F1 (Macro)	Unweighted average of per-class F1 scores	Equal weight to all categories
Precision (Weighted)	Weighted average of per-class precision	Controls false positive rate
Recall (Weighted)	Weighted average of per-class recall	Controls false negative rate
Top-3 Accuracy	Correct class in top 3 predictions	Captures category overlap ambiguity

7.2 Results Comparison

Metric	Baseline (V1)	Advanced (V2)	Improvement
Categories	57	56	Collision resolved
Accuracy	68.16%	72.01%	+3.85%
F1 (Weighted)	67.68%	71.77%	+4.09%
F1 (Macro)	65.31%	69.03%	+3.72%
Precision (Weighted)	67.72%	71.80%	+4.08%
Recall (Weighted)	68.16%	72.01%	+3.85%
Top-3 Accuracy	88.29%	92.38%	+4.09%
Training time	~102 minutes	~111 minutes	Comparable

The advanced model achieves consistent improvements of approximately 4% across all metrics. The 92.38% Top-3 accuracy demonstrates that the model's understanding of the domain is substantially deeper than the Top-1 accuracy suggests, with the correct category appearing among the top three predictions for over 92% of test samples.

7.3 Training Dynamics

The training curves for the advanced model show healthy convergence behavior. The training loss decreases steadily from approximately 4.0 to 1.1 over the course of training. The validation loss converges to approximately 1.25, with the gap between training and validation loss indicating well-controlled overfitting. Validation accuracy and F1 score both show consistent improvement with diminishing returns after step 5,000, confirming that the model was appropriately trained to convergence.

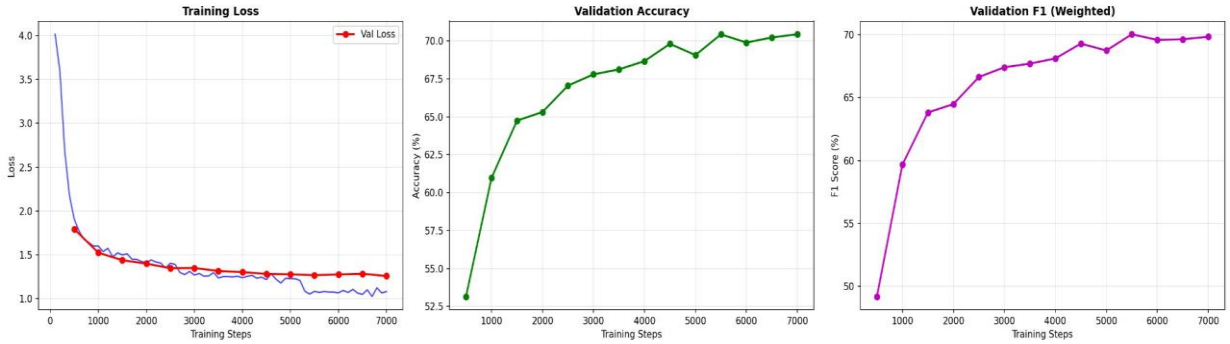

Figure 6: Training curves for the advanced model showing loss (left), validation accuracy (center), and validation F1 (right).

7.4 Confusion Matrix Analysis

The confusion matrix of the advanced model reveals dramatically cleaner classification compared to the baseline. High-energy physics experimental (hep-ph) achieves 91% recall, up from 81% in the baseline. Astrophysics cosmology (astro-ph.CO) reaches 92% recall, up from 65% in the baseline. Notably, the parent-child collisions that plagued the baseline (astro-ph being confused with astro-ph.CO at 36%) are completely eliminated because the collision parent was resolved in preprocessing.

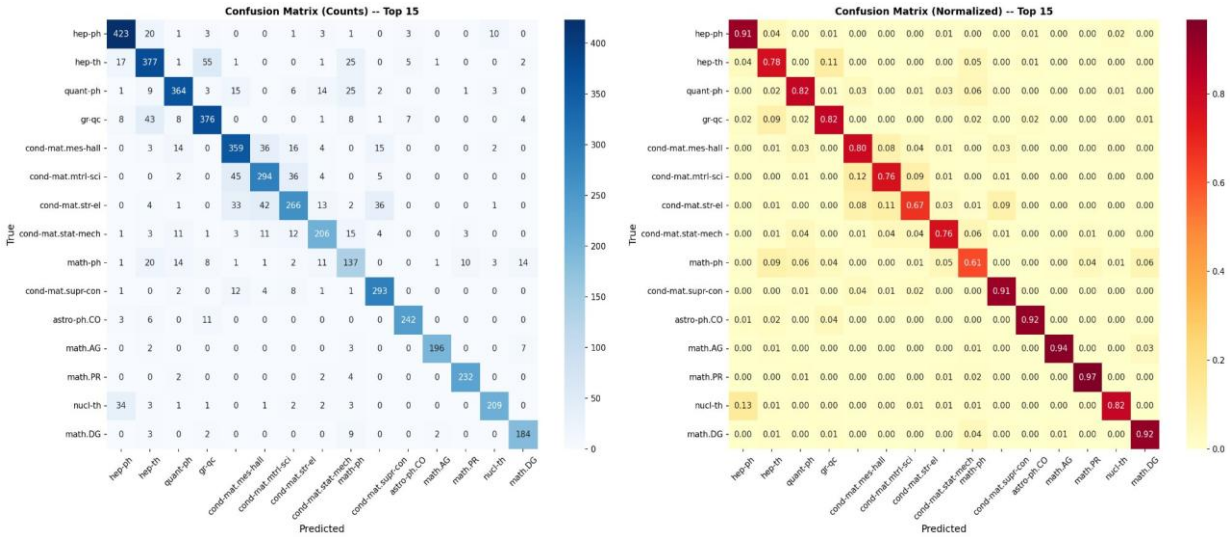

Figure 7: Confusion matrix for the advanced model. Raw counts (left) and normalized proportions (right) for top 15 categories.

7.5 Per-Class Performance

The per-class F1 analysis reveals a spectrum of classification difficulty. The best-performing categories are cs.IT (Information Theory, F1=0.875), hep-lat (Lattice, F1=0.873), and hep-ex (Experiment, F1=0.864), all of which have highly distinctive vocabulary. The most challenging categories are nlin.SI (Exactly Solvable, F1=0.386), math-ph (Mathematical Physics, F1=0.428), and cond-mat.other (F1=0.434), which overlap semantically with multiple other categories. The mean per-class F1 is 0.671, indicating that the model performs substantially above chance (1/56 = 1.8%) across the full range of categories.

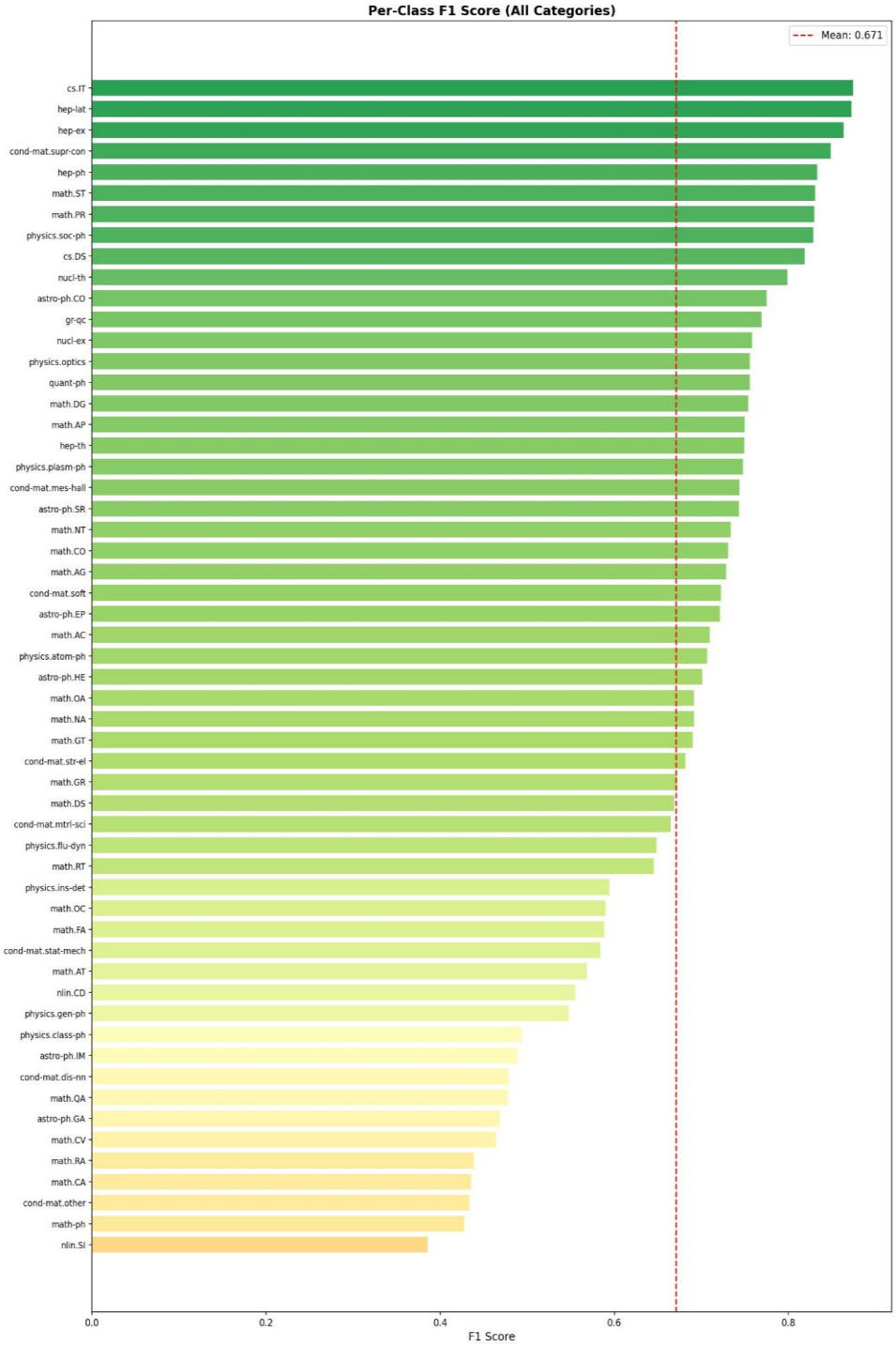

Figure 8: Per-class F1 scores for all 56 categories, sorted by performance. Red dashed line indicates the mean (0.671).

8. Hyperparameter Tuning

We conducted a structured hyperparameter search to identify the optimal configuration for our advanced model. Based on the architecture's novel components, we focused on three parameters with the highest expected impact: Focal Loss gamma (which controls hard-example focusing), the number of pooling layers, and the learning rate.

8.1 Search Configuration

To keep runtime manageable, we screened six configurations on a 20,000-sample training subset and 5,000-sample validation subset, each trained for 3 epochs. The best configuration was then retrained on the full training set for 5 epochs with early stopping.

Configuration	Learning Rate	Focal Gamma	Pooling Layers	F1 (Weighted)
gamma=1.0 (mild focal)	2e-5	1.0	4	60.90% (best)
gamma=2.0 (standard focal)	2e-5	2.0	4	58.41%
gamma=3.0 (aggressive focal)	2e-5	3.0	4	55.04%
LR=3e-5	3e-5	2.0	4	59.00%
2-layer pool	2e-5	2.0	2	58.34%
1-layer pool (CLS only)	2e-5	2.0	1	57.27%

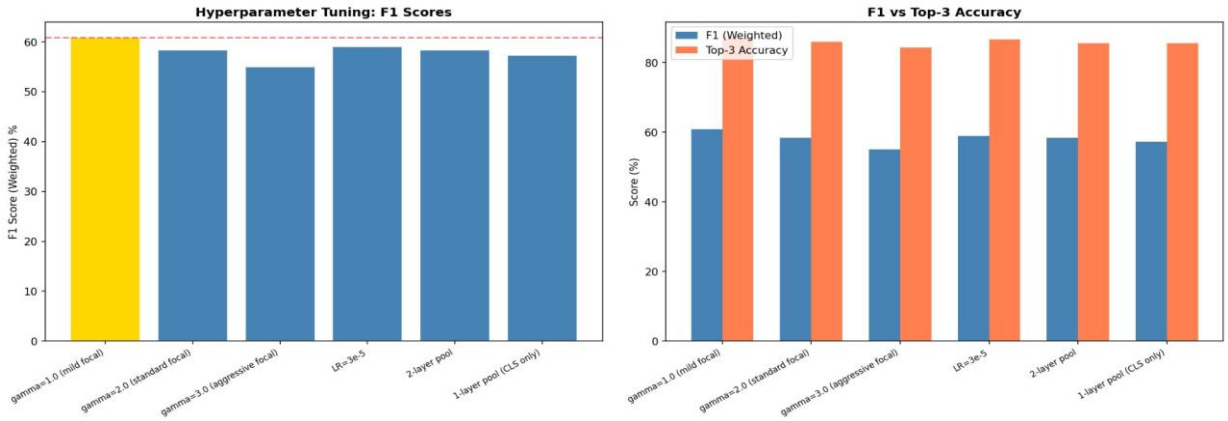

Figure 9: Hyperparameter tuning results. F1 scores (left) and F1 versus Top-3 accuracy (right).

8.2 Key Findings from Tuning

The hyperparameter search yielded several important insights. First, a mild focal loss (gamma=1.0) outperformed both standard (gamma=2.0) and aggressive (gamma=3.0) settings, suggesting that for this dataset, moderate refocusing is more beneficial than extreme hard-example mining. Second, 4-layer pooling consistently outperformed 2-layer and 1-layer (CLS-only) configurations, confirming the value of multi-layer representations for scientific text. Third, the learning rate of 2e-5 remained optimal, confirming the recommendation of Sun et al. (2019). The best

configuration (gamma=1.0, 4-layer pooling, LR=2e-5) achieved 60.90% F1 on the screening subset and 71.77% when retrained on the full dataset.

9. Optimal Number of Categories

Determining the optimal number of categories requires balancing three competing objectives: (1) granularity (more categories provide finer-grained classification), (2) coverage (more papers should be classifiable), and (3) learnability (each category needs sufficient training data). Our analysis operates at three granularity levels.

At the major category level (18 categories), classification would be straightforward but loses the meaningful distinctions between subcategories. At the full primary category level (141 categories), many classes have insufficient samples for reliable learning. Our recommended level uses primary categories filtered to a minimum of 500 samples each, followed by hierarchical resolution of parent-child collisions. This yields 56 categories covering over 93% of the dataset, with each category having enough training examples for the model to learn robust decision boundaries.

The coverage-versus-granularity trade-off curve (Figure 5) confirms that this threshold represents an optimal balance: below 500 samples, model performance degrades significantly for rare categories, while above 1,000 samples, too many meaningful categories are lost.

10. Conclusions and Findings

This work presents a complete, end-to-end NLP pipeline for classifying arXiv research articles into subject categories using their abstracts. Starting from a standard SciBERT baseline (68.16% accuracy), we systematically identified weaknesses and developed targeted solutions, achieving a final performance of 72.01% accuracy, 71.77% weighted F1, and 92.38% Top-3 accuracy across 56 categories.

10.1 Key Findings

Our first and most impactful finding is that parent-child category collisions in hierarchical taxonomies represent a fundamental, irreducible source of classification error. The collision between astro-ph and its children (astro-ph.CO, astro-ph.GA, and others) alone accounted for approximately 5% of baseline errors. No amount of model tuning can overcome this data-level ambiguity; it must be resolved in preprocessing. This insight generalizes to any classification task with hierarchical label structures.

Second, Focal Loss with mild focusing (gamma=1.0) provided the most consistent per-class improvement across the full category distribution. Unlike standard cross-entropy, which allows the model to achieve low loss by excelling only on common categories, Focal Loss redirects gradient signal to difficult and underrepresented classes.

Third, multi-layer [CLS] pooling captures complementary information across transformer depth levels. Our ablation study confirmed that 4-layer pooling consistently outperforms 2-layer and single-layer configurations, with lower layers contributing syntactic discrimination (useful for

distinguishing mathematical formalism from experimental methodology) and upper layers contributing semantic topic information.

Fourth, the overlap between semantically related categories represents genuine research interdisciplinarity rather than model failure. Our Top-3 accuracy of 92.38% confirms that the correct category almost always appears among the model's top predictions, even when the Top-1 prediction selects a closely related alternative.

Fifth, layer-wise learning rate decay proved essential for stable fine-tuning of the multi-layer architecture. Without it, the classification head's large gradients destabilize the pre-trained transformer weights, leading to degraded performance.

10.2 Recommendations for Future Work

Several promising directions exist for extending this work. Multi-label classification using binary cross-entropy with logits loss would better capture papers that genuinely span multiple fields. Hierarchical classification using a two-stage cascade (predict major category first, then subcategory) could exploit the natural taxonomy more effectively. Continued pre-training of SciBERT on recent arXiv papers from 2022 to 2025 would address vocabulary drift as new research areas emerge. Finally, embedding-based clustering using UMAP dimensionality reduction and HDBSCAN could discover latent category structures that may differ from the official arXiv taxonomy.

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